

Spatial Analysis and Modeling — Intro

Why location changes data, inference, and models

Spatial Analysis and Modeling · University of Turin



**Why spatial thinking is
important?**

What spatial analysis and modeling studies

Spatial analysis studies data where **location is part of the evidence**, not only a label for display. The central question is whether spatial arrangement changes what we can infer from the data. Spatial modeling turns that question into formal structure:

- **Represent** observations as points, areas, fields, or networks.
- **Measure** spatial pattern, dependence, scale, and uncertainty.
- **Model** processes whose behavior varies across space.
- **Evaluate** predictions under spatial sampling and transfer.

The course moves from maps as descriptions to spatial models as evidence.

When a question is spatial

A research question becomes spatial when changing the location changes the answer.

Descriptive

- Where are events concentrated?
- Which areas are similar?
- How does a surface vary?

Explanatory or predictive

- Why does risk cluster here?
- Do nearby units influence each other?
- Will a model trained here work elsewhere?

Spatial analysis determines whether spatial structure changes measurement, inference, or prediction.

It started with a map

The table recorded deaths. The map revealed a spatial pattern that could be linked to a mechanism: contaminated water supply. Clustered around the Broad Street pump.



What is geography?

Before spatial analysis there is geography: the discipline that studies how places, environments, and human activities vary across the Earth's surface.

Physical geography

The natural surface: terrain, climate, rivers, soils, ecosystems — the processes that shape where things *can* happen.

Spatial thinking adds two questions to ordinary data analysis: **where is it?** and **why there?**

Human geography

People on that surface: cities, mobility, health, economies, culture — the processes that shape where things *do* happen.

Geography is about *processes*, not just places

Spatial data are usually observations of a process: disease transmission, housing markets, migration, land-cover change, pollution transport, or urban accessibility.

A place is not only a coordinate. It is a position inside a system of neighbours, flows, barriers, boundaries, institutions, and measurement choices.

Professional question: what process could have produced the observed spatial pattern, and what evidence would distinguish it from alternatives?

Two different questions

Spatial work often separates description from explanation. Both matter, but they answer different questions.

Describing *where*

Mapping & GIS. Put things in their correct location, draw the map, measure distances and areas.

Where are the cholera deaths?

A map shows spatial arrangement. Spatial analysis asks whether the arrangement is stronger than expected, whether it changes inference, and which mechanisms are plausible.

Explaining *why here*

Spatial analysis and modeling. Test whether the pattern is systematic, estimate relationships, and assess uncertainty.

Why do the deaths cluster around that one pump?

What makes spatial data special?

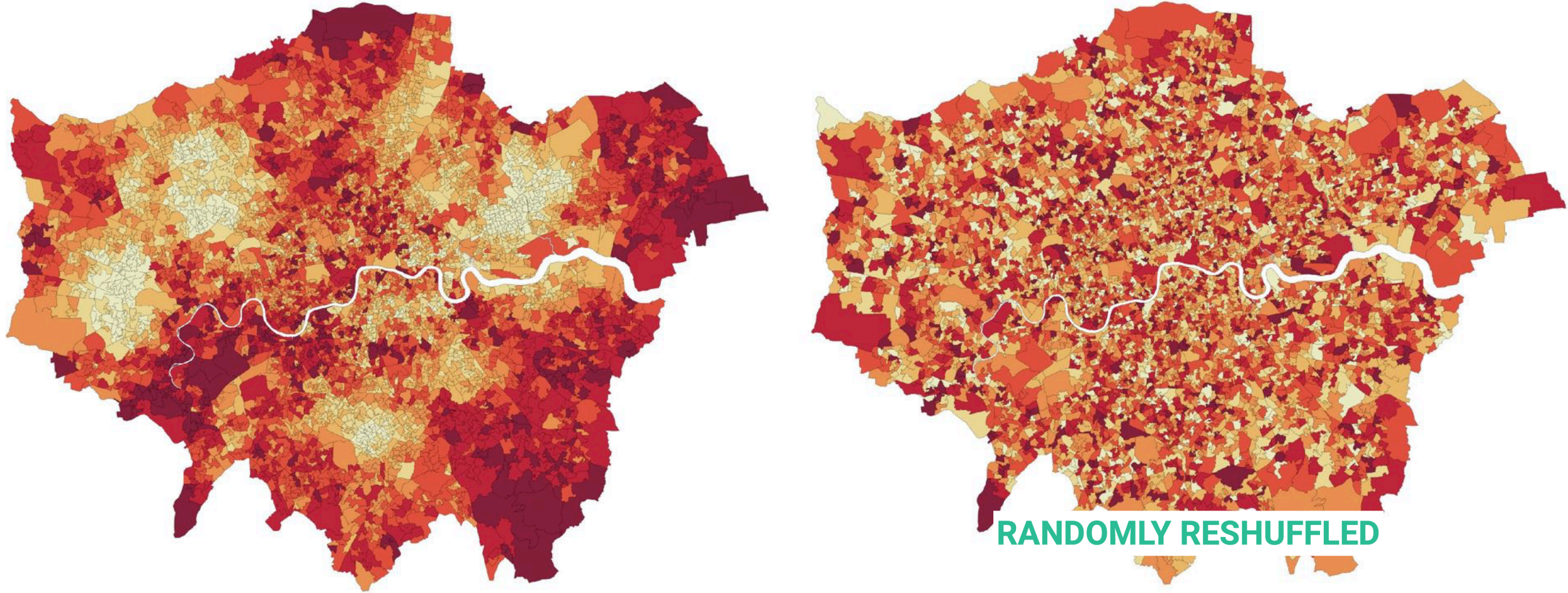
Once observations have locations, two issues appear that ordinary tables usually hide.

- **Spatial dependence** — nearby observations often share information because they are connected by exposure, movement, diffusion, markets, or measurement design.
- **Spatial heterogeneity** — relationships can change across regions, scales, directions, or institutional contexts.

These two ideas organize much of the course. Scale, boundaries, sparse labels, networks, and spatial transfer all interact with dependence and heterogeneity.

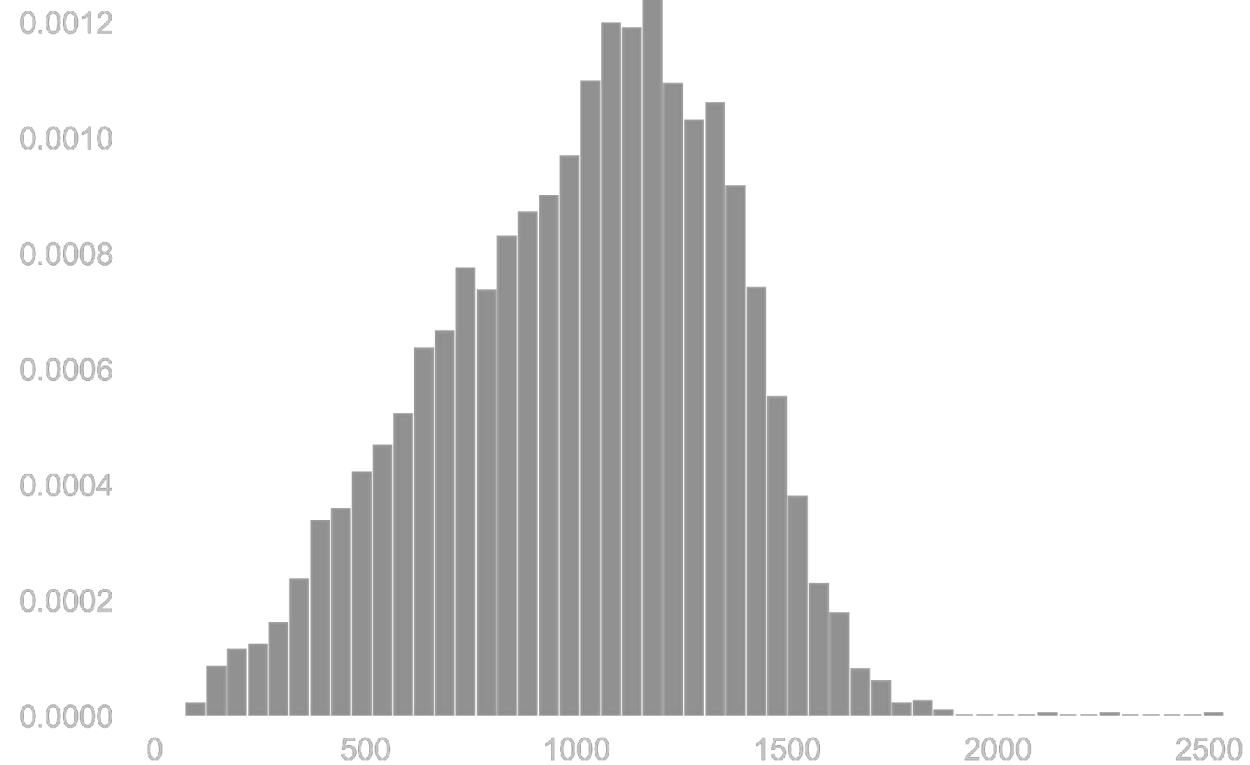
Same data, two very different maps

Left: percentage of white residents in London (LSOA, census 2011). **Right:** the *same values* reshuffled to random locations



Same values, same histogram

Same distribution; different spatial arrangement.



The First Law of Geography

Tobler's First Law is a compact statement of spatial dependence. It is not a model by itself; it is a warning that independence is often the wrong starting assumption.

"Everything is related to everything else, but near things are more related than distant things."

— Waldo Tobler, 1970

The rest of the course turns this intuition into measurable choices: how to define neighbours, how to test autocorrelation, how to model spillovers or residual structure, and how to validate predictions away from observed locations.

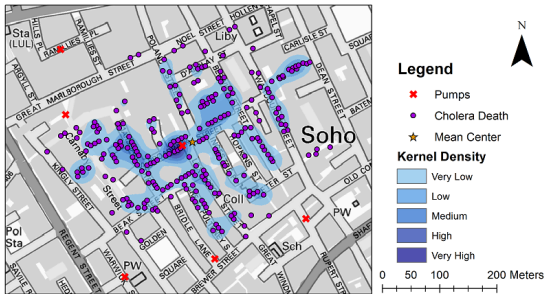
Three lenses on the world

Before choosing a method, choose a representation: the same phenomenon can be objects, fields, or networks — each supporting different operations.

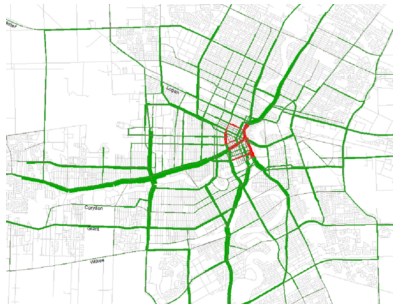
Objects — discrete features

Things with crisp edges you point at and count.

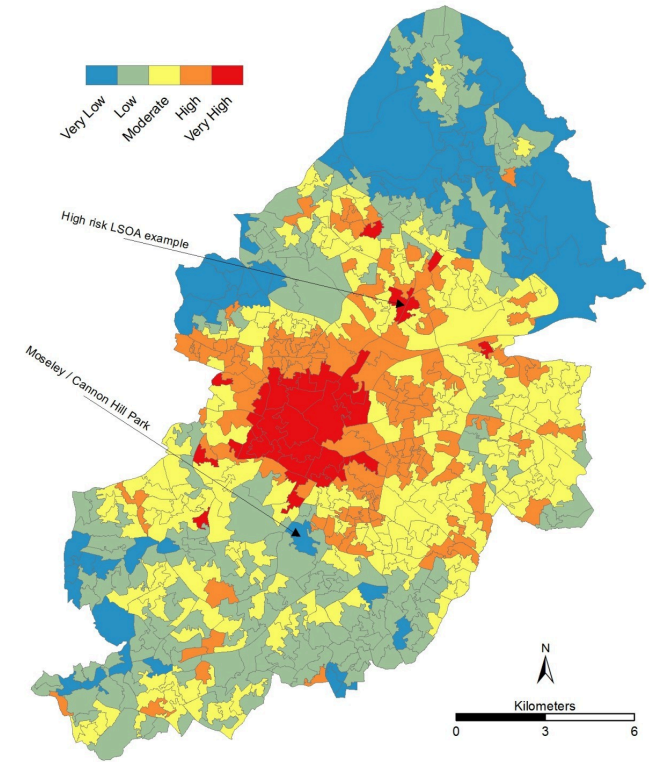
- **Points** : locations (trees, accidents)



- **Lines** : paths (roads, rivers)



- **Polygons** : areas (city boundaries, lakes)



Fields and networks

A field assigns values over continuous space. A network routes interaction along links rather than through straight-line distance.

Fields — continuous surfaces

A value *everywhere*, not just where an object sits — think elevation or temperature, defined at every location and sampled at points or on a regular grid.

A city, three lenses: clinics = **objects**, air quality = **field**, streets = **network**.

Networks — connected flows

Reality where movement follows **links**, not straight lines: roads, rivers, transit, social ties. "Distance" becomes *travel along the network*, and rivers, borders, and roads channel how things interact.

You meet this everywhere

Earth Science

- Land use/cover mapping using satellite imagery
- Deforestation monitoring
- Disaster mapping (floods, earthquakes, hurricanes)
- Species distribution modeling
- Soil property mapping

Urban Informatics

- Traffic prediction and routing
- Crime hotspot prediction
- Air quality mapping
- Ride-sharing demand forecasting

Public Health

- Disease risk mapping
- Outbreak detection and monitoring
- Environmental health factor analysis
- Drug epidemic understanding

Geosocial Media Analytics

- Real-time event detection from geotagged social media posts
- Political unrest forecasting
- Travel recommendation systems

Different fields, same two problems: dependence and heterogeneity.

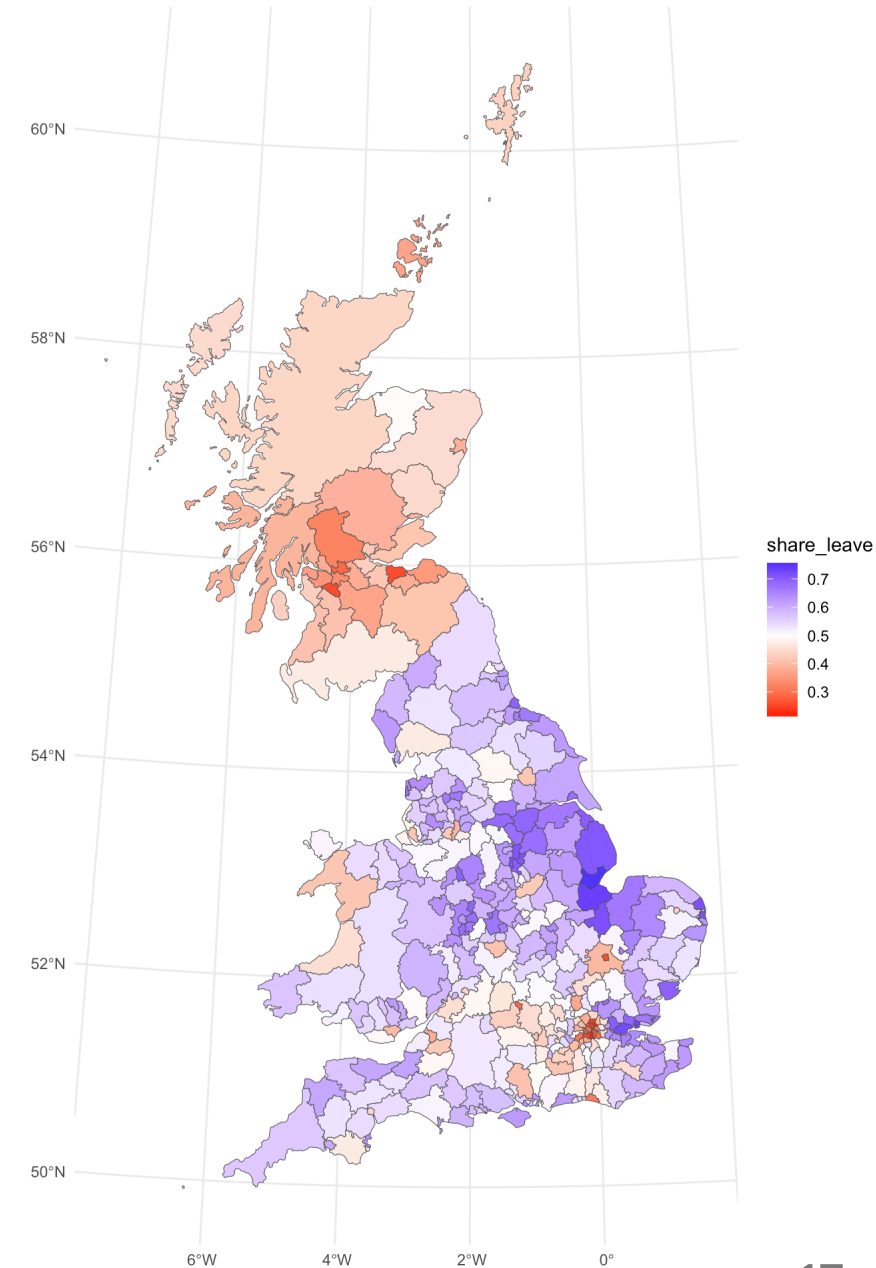
When dependence bites

Nearby observations may not provide independent evidence. Many standard estimators and validation schemes assume independent errors or independent samples; spatial structure can violate those assumptions.

What goes wrong

- **Regression:** residual errors are correlated, not i.i.d.
- **Classification:** salt-and-pepper noise across the map
- **Inference:** the independence assumption is simply false

Example: in an EU-referendum *leave*-share map, neighbouring areas tend to vote similarly. A model that assumes independent observations treats that clustering as accidental rather than structured.



When the rules change across the map

Spatial heterogeneity means that one relationship may not describe the whole study area. A coefficient estimated globally can hide different local regimes.

Non-stationarity

- The relationship varies across sub-regions
 - same spectral signature → different land cover (tropical vs. temperate)
- **Spatial regimes:** a single global model can fail locally

Anisotropy

- Dependence can vary by **direction**
 - climate shaped by mountain ranges
 - water flow follows terrain and gravity

Implication: a single global model may be a poor summary when mechanisms vary by region, direction, or scale.

When scale changes the answer

Spatial data are observed and reported at many resolutions. Aggregating the same underlying observations into different units can change the estimated relationship.

The problem

- **Earth imagery:** sub-metre (aerial) to 250 m–1 km (MODIS satellite)
- **Agriculture:** point sensors + plot-level yields + aerial imagery
- **Aggregation:** combine the same data into different units and the relationship can shift — or flip

Implications

- Models assume a single scale/resolution (violated)
- Simple aggregation loses critical information

Maps are analytical models

A map is a model: it selects a projection, boundary system, classification, and scale. The **Modifiable Areal Unit Problem (MAUP)** shows why those choices affect analysis. The same individual observations can yield different aggregate patterns under different spatial units.

- **Zonal effect** — same number of units, different boundary configuration
 - keep every vote, redraw the lines (the gerrymander on the next slide)
- **Scale effect** — fewer, larger units (US counties versus states; zip codes versus census tracts)
 - coarser aggregation can strengthen, weaken, or even flip a correlation
- See *Openshaw & Taylor (1979), "A million or so correlation coefficients."*

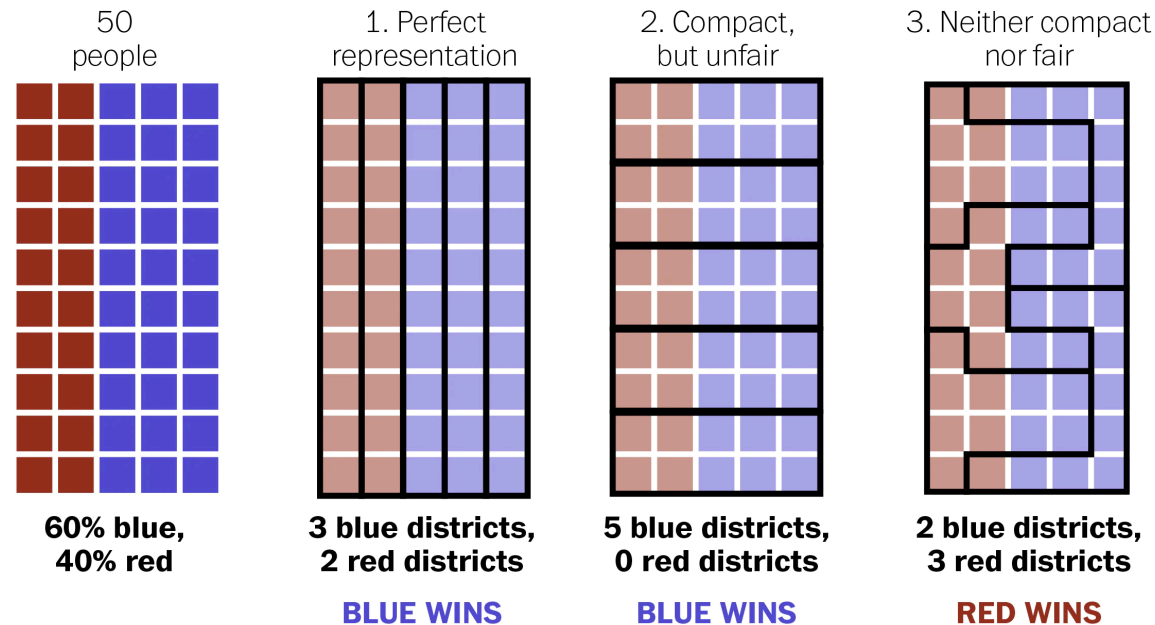
Analytical consequence: spatial units are part of the research design. Report them, justify them, and test whether conclusions depend on them.

Same voters, different boundaries

Gerrymandering is the MAUP turned into a weapon: keep every vote unchanged, redraw the lines, and flip the outcome. Proof that *where you put the boundary* is itself a decision.

Gerrymandering, explained

Three different ways to divide 50 people into five districts



The ecological fallacy

One consequence of aggregation deserves its own name, because it misleads real policy. The **ecological fallacy** is assuming that a relationship measured between variables at an *aggregate* level (areas, counties) also holds for the *individuals* inside them.

- The group-level association can differ in **sign or magnitude** from the individual-level one.
- **Aggregation hides within-group variability** and **individual heterogeneity**.
- Confounding and composition effects within units can produce spurious group-level correlations.
- Simpson's paradox — **aggregated and disaggregated trends can point in opposite directions**.

Rule: do not make individual-level causal claims from area-level coefficients. The result may describe places, not people.

Robinson (1950), *Ecological correlations and the behavior of individuals*, American Sociological Review 15(3):351–357.

A preview of the toolkit

Aggregation problems cannot be solved by better wording. They require a match between the question, the data level, and the model.

- **Match the inference target to the data level:** aggregate data → aggregate conclusions; individual claims need individual data.
- **Multilevel / hierarchical models** (partial pooling) when individual and area data are mixed.
- **Finer-scale estimation** — small-area estimation, downscaling — with explicit uncertainty.

When the labels are costly

Spatial data collection is often expensive because locations must be visited, interpreted, or measured. The design question is therefore not only sample size, but spatial coverage.

Why it's different

- **High collection costs:** field crews, travel time between locations
- **Labeling costs:** visual interpretation by experts
- **Geographical representativeness:** samples must be spatially balanced, not just numerous

What it pushes us toward

- Semi-supervised methods become critical
- Sampling has to be *spatially* aware — plain active learning ignores travel cost

How the course answers back

The course links each spatial problem to a family of methods and a diagnostic habit.

- **Spatial dependence** : contextual features, dependency baked into the model, regularization
- **Spatial heterogeneity** : location-dependent models, ensembles, multi-task learning
- **Limited ground truth** : semi-supervised, active learning
- **Multiple scales** : spatial hierarchical models

(This course focuses on the first two.)

One line of notation to carry forward

Most methods in the course can be written as supervised learning or regression with locations attached to observations.

Given samples $\{(x(s_i), y(s_i))\}_{i=1}^n$ at locations s_i , learn f such that

- Regression: $y(s) \approx f(x(s))$
- Classification: $\hat{y}(s) = f(x(s)) \in \mathcal{C}$
 - where $\mathcal{C}$ is the set of possible classes (e.g. land-cover types)

The spatial twist: observations are **not i.i.d. (independent & identically distributed)** — dependence and heterogeneity are the rule, not the exception.

Learning objectives: foundations

By the end of this class, you will be able to:

- Recognize the unique challenges of spatial data (dependence, heterogeneity, scale, anisotropy).
- Choose an appropriate spatial representation and CRS for measurement, joining, and modeling.
- Use GeoPandas to implement vector-data cleaning, feature engineering, and exploratory analysis in Python.
- Decide when files, GeoPandas, PostGIS, DuckDB, GeoParquet, or distributed tools fit the research workflow.

Learning objectives: modeling

By the end of this class, you will be able to:

- Formalize a spatial prediction task with clear notation and assumptions.
- Build spatial weights and compute and interpret Moran's I and LISA, variograms, and point-pattern statistics.
- Choose among SAR/SEM/SDM spatial econometric models, with an introduction to kriging.
- Model spatial heterogeneity with GWR/MGWR.
- Design spatial cross-validation (blocking/buffering) and diagnose residual spatial autocorrelation.
- Apply spatial clustering and regionalization, and communicate method trade-offs.

Course roadmap

The course moves from spatial foundations to data management, modeling, validation, and regionalization.

- **01 · Intro** — why spatial is special: dependence, heterogeneity, Tobler
- **02 · Spatial data** — CRS, GeoPandas, geometries, vector/raster/network data, and formats
- **03 · Spatial databases** — GeoPandas with PostGIS, DuckDB, GeoParquet, indexing, and spatial joins
- **04 · Geovisualization** — turning spatial data into maps that tell the truth
- **05 · Spatial dependence** — making Tobler measurable: weights, Moran's I , LISA, variograms & kriging, point patterns, spatial econometrics (SAR/SEM/SDM)
- **06 · Modeling heterogeneity** — when the rules change across space: GWR / MGWR
- **07 · Evaluation** — spatial cross-validation, diagnostics, uncertainty
- **08 · Clustering & regionalization** — grouping places into meaningful regions

The sequence is cumulative: later modeling choices depend on representation, CRS, spatial data management, and evaluation design.